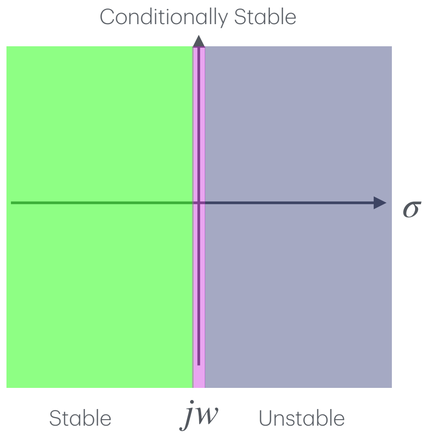
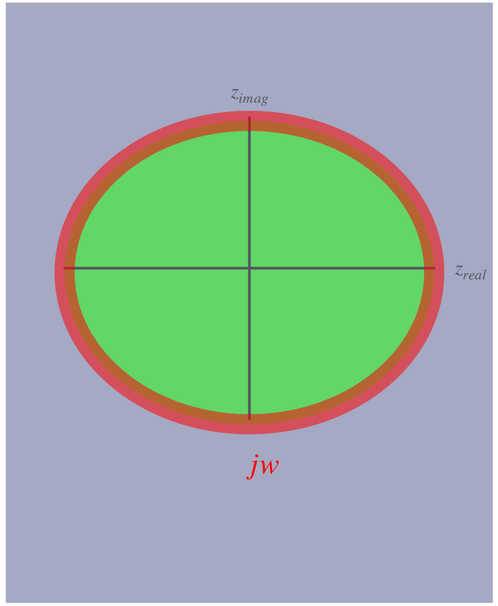
Z - Tranform

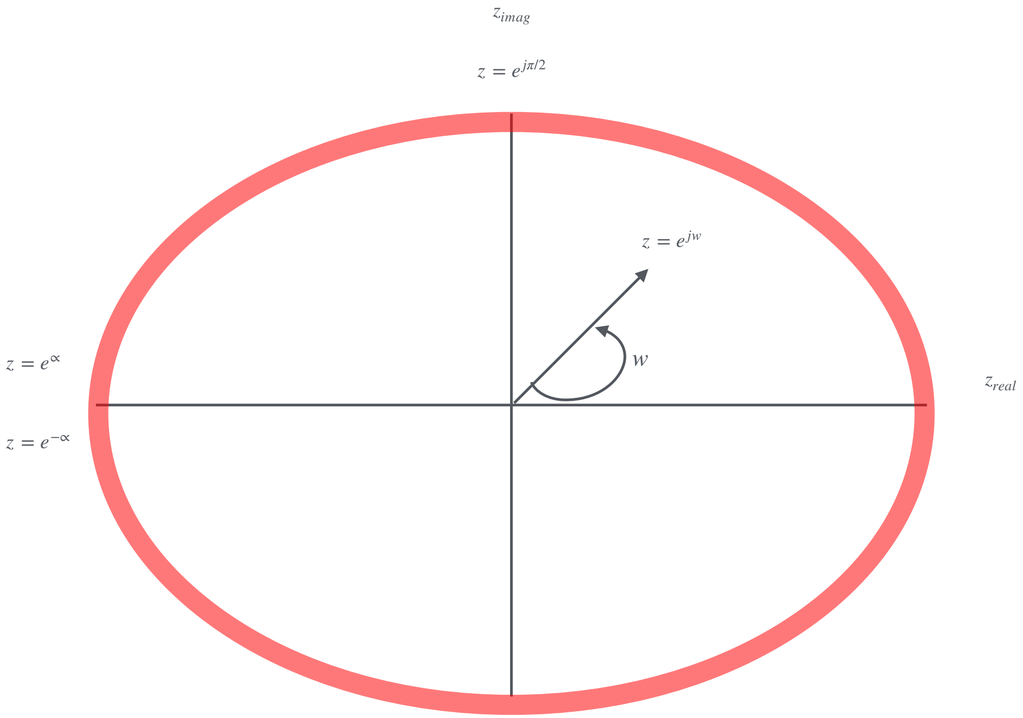
HW



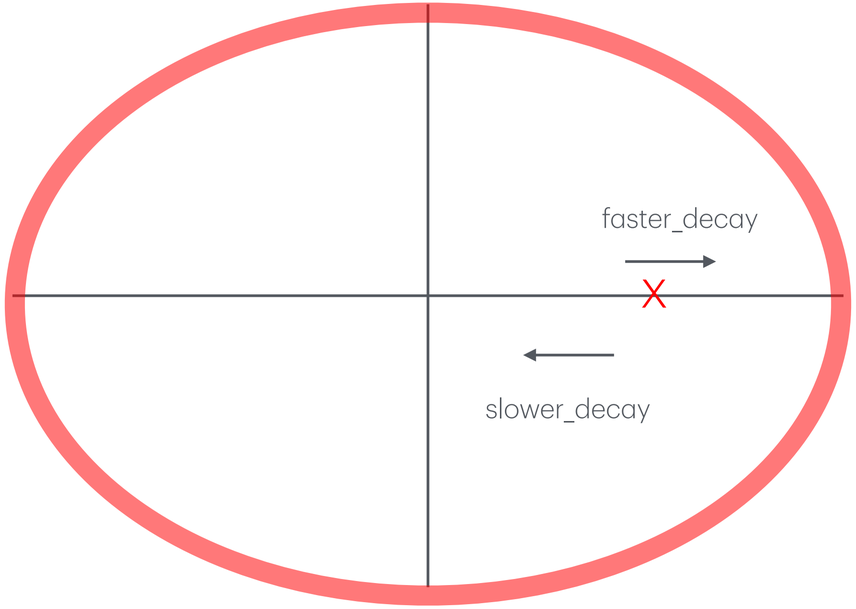
Laplace Stability map



Z-Transform Stability map


z_{imag}
$z = e^{j\pi/2}$
$z = e^{jw}$
w
$z = e^{\alpha}$
$z = e^{-\alpha}$
z_{real}

Poles at certain location generate impulse responses

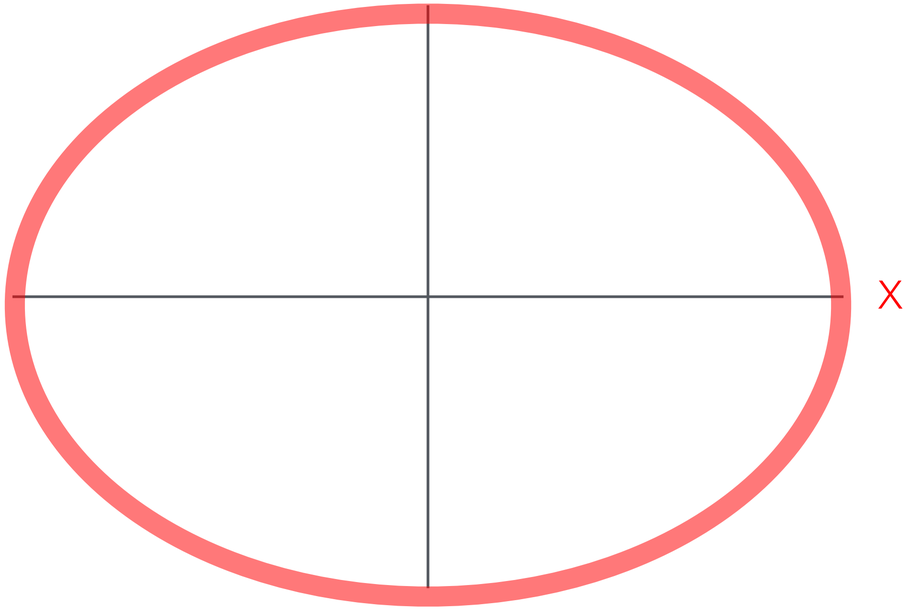


*											
	*										
		*									
			*								
				*	*	*	*	*	*	*	*

$$\frac{1}{1 - pz^{-1}} \quad p < 1$$

notice convergence to zero

Poles at certain location generate impulse responses



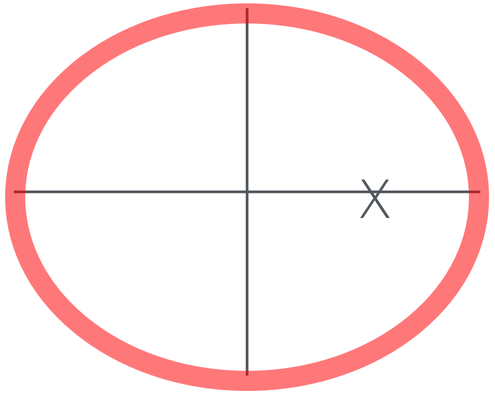
								*	*	*	*
							*				
						*					
					*						
*	*	*	*	*							

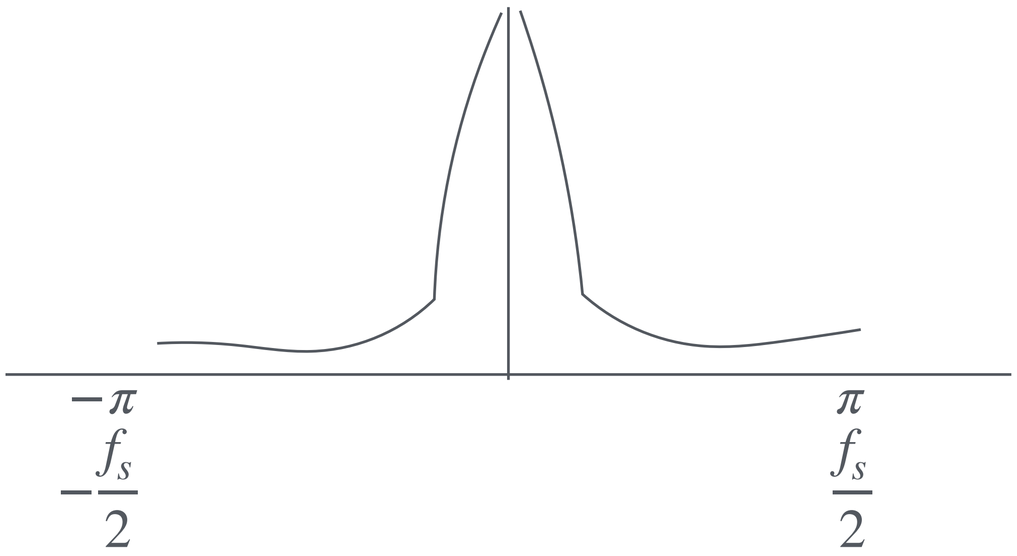
$$\frac{1}{1 - pz^{-1}} \quad p > 1$$

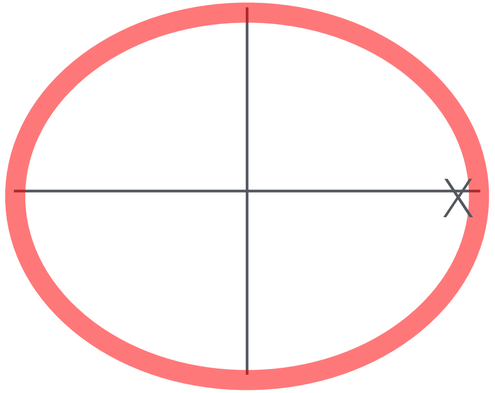
Impulse response does not converge to zero (unstable system)

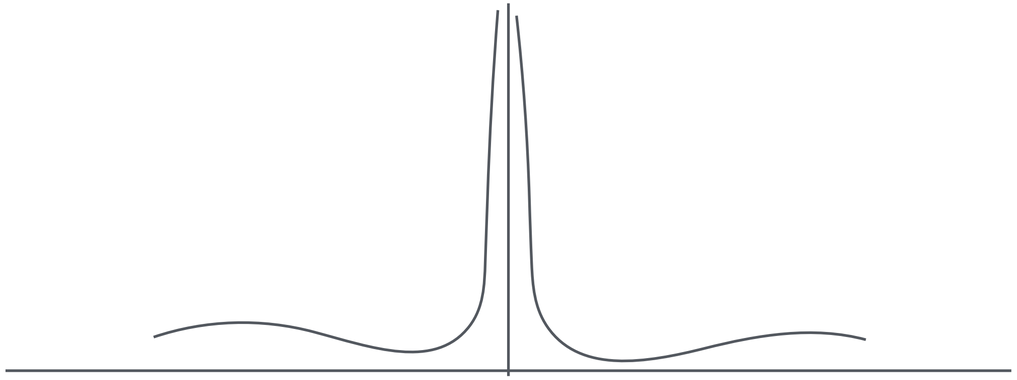
IIR Filter

- Improved Frequency Response with weak side-lobes
- Less computation (fewer hardware resources)
- Nonlinear phase response
  - h(n) not symmetric

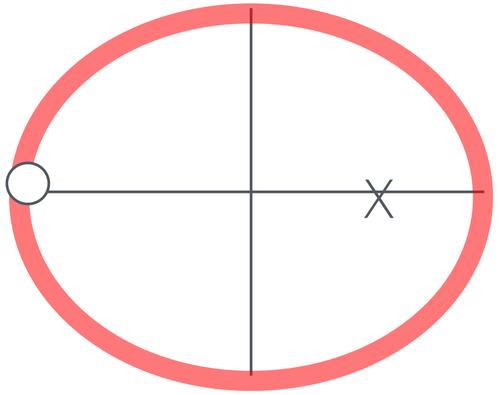


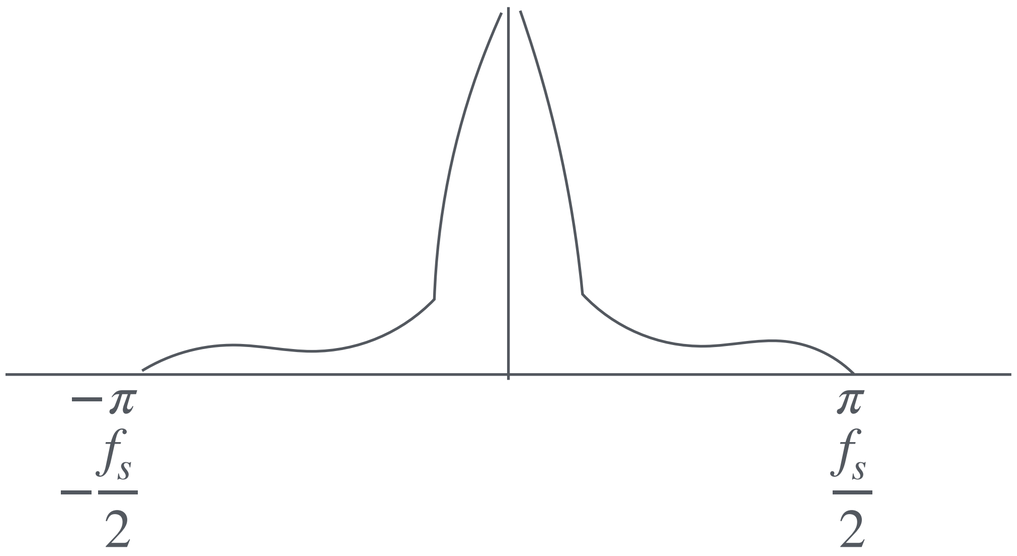

$-\pi$
$-\frac{f_s}{2}$
π
$\frac{f_s}{2}$



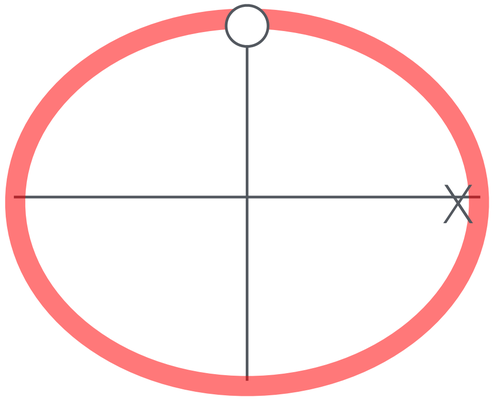


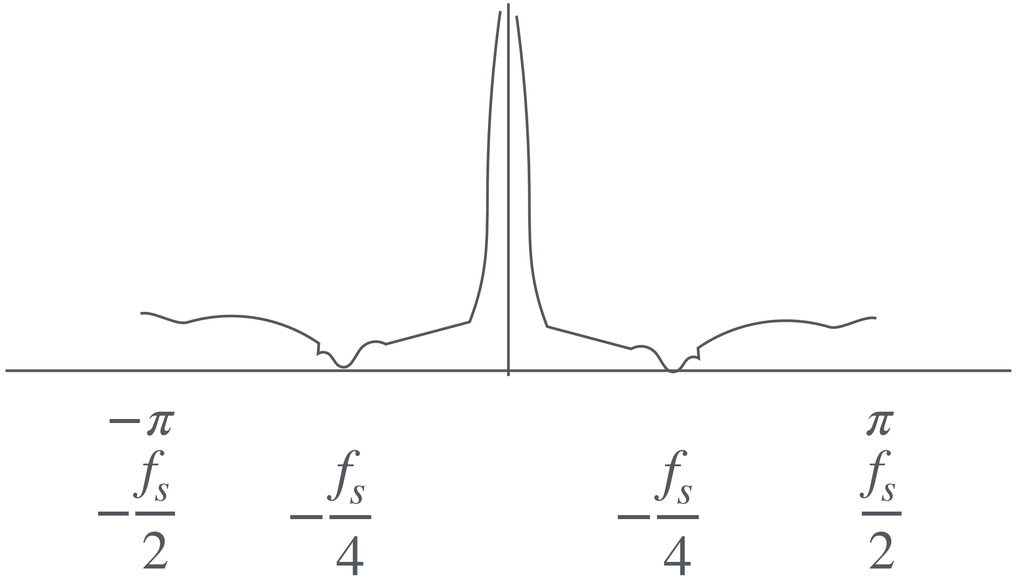
IIR FILTER



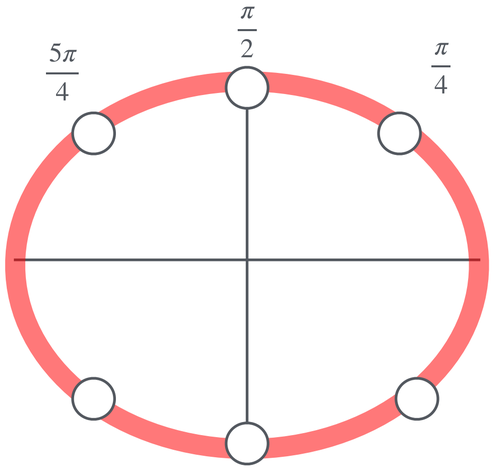


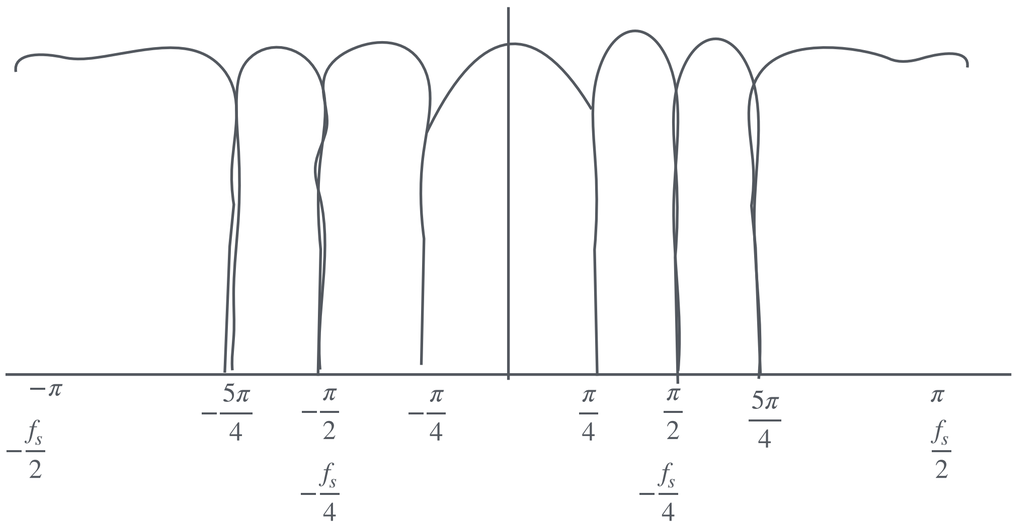
IIR FILTER





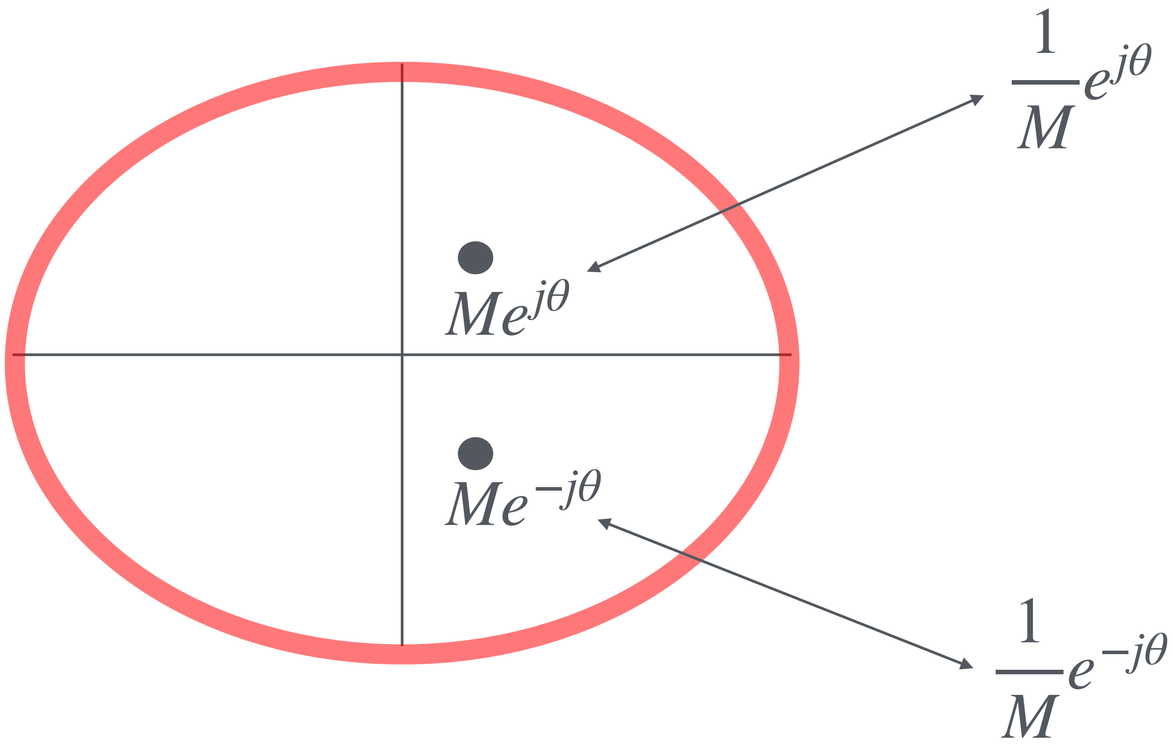
FIR FILTER


$\frac{\pi}{2}$
$\frac{5\pi}{4}$
$\frac{\pi}{4}$


$-\pi$
$-\frac{f_s}{2}$
$-\frac{5\pi}{4}$
$-\frac{\pi}{2}$
$-\frac{f_s}{4}$
$-\frac{\pi}{4}$
$\frac{\pi}{4}$
$\frac{\pi}{2}$
$\frac{f_s}{4}$
$\frac{5\pi}{4}$
π
$\frac{f_s}{2}$

FIR Filter Linear Phase

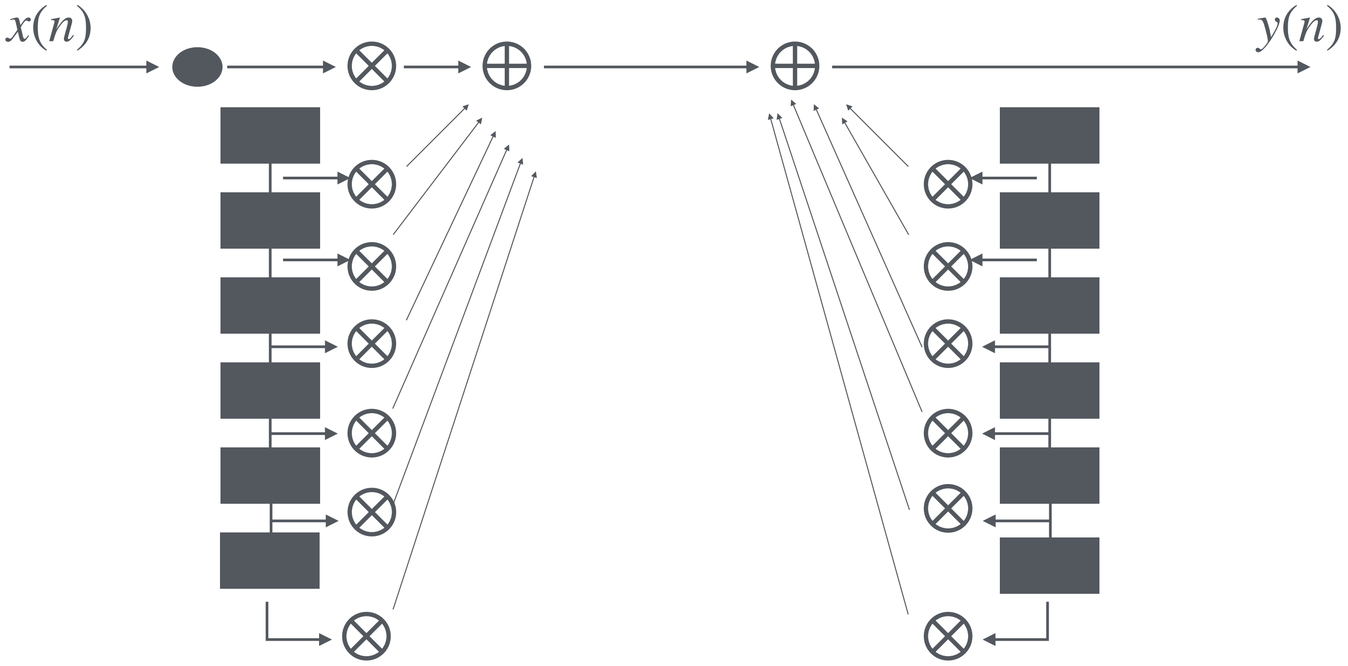
- Zeros without a magnitude 1 requires accompanying zero with amplitude equal to 1/M and negative of the phase



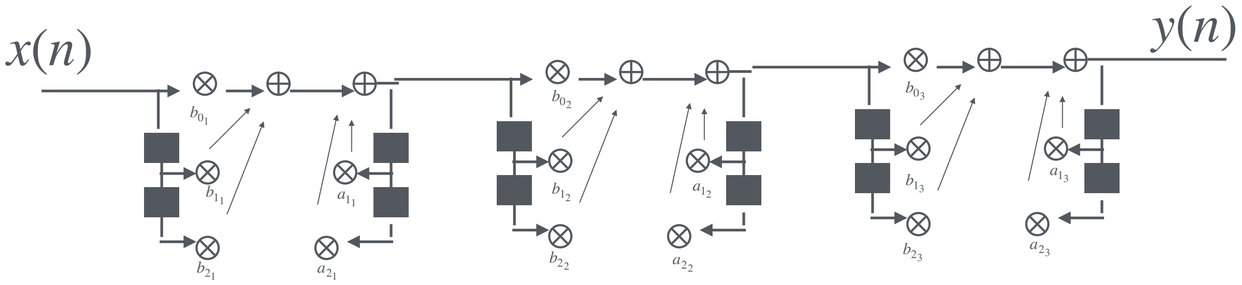
IIR Structures

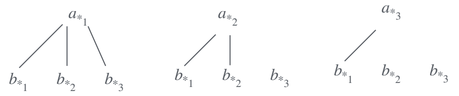
- Direct Form I
  - Resistant to coefficient quantization and stability problems
- Direct Form II
- Transposed Direct Form II
  - Best performance on accumulators

Cascading Filters



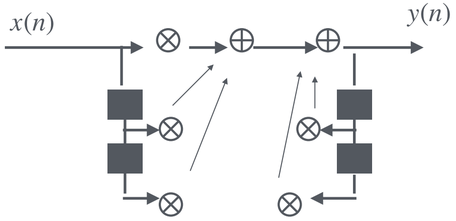
Cascading Filters





6 possible coefficient pairing using 3 sets a, b coefficients

3! possible permutations or ordering of sub filters



DC Gain $f = 0$:

$$\left(\frac{Y(f=0)}{X(f=0)}\right)$$

Unity Gain Lowpass Filter:

Divide filter feedforward coefficient by DC gain (frequency gain response at f = 0)

DC Gain $f_s/2$:

$$\left(\frac{Y(f=\frac{f_s}{2})}{X(f=\frac{f_s}{2})}\right)$$

Unity Gain Highpass Filter:

Divide filter feedforward coefficient by frequency magnitude response at $\frac{f_s}{2}$